

**UNIVERSITY OF DHAKA**

# **AI-Driven Real-Time Detection of Fake News and Malicious Content**

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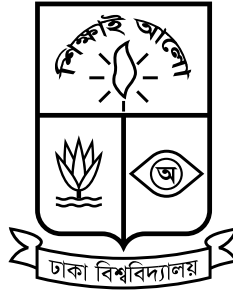
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# Declaration Form



University of Dhaka

We hereby declare that the work in this report titled "AI-Driven Real-Time Detection of Fake News and Malicious Content" has been carried out by us under the supervision of Mr Jargis Ahmed and co-supervision of Mr Md. Faisal Hossain. We further declare that where information has been derived from the literature, we have adequately cited and referenced the original sources. No part of this report was previously presented for another degree or diploma at this or any other institution.

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*“Common sense is like deodorant. The people who need it most never use it.”*

Bill Murray

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*Abstract*

The rapid growth of digital platforms and social media has significantly accelerated the spread of fake news and malicious content, posing serious challenges to information reliability and cybersecurity. Traditional detection approaches are often limited by delayed response, lack of contextual understanding, and inability to handle real-time data. To address these issues, this project proposes an AI-driven real-time detection system capable of identifying and classifying content as real, fake, or malicious. The proposed system utilizes datasets collected from publicly available sources containing labeled examples of news and malicious text. The data is preprocessed through cleaning, normalization, and tokenization to ensure quality input. Feature extraction is performed using both TF-IDF and BERT embeddings, enabling the system to capture both statistical patterns and contextual meaning of text. For classification, multiple machine learning models such as Logistic Regression, Naive Bayes, and Random Forest are implemented, while a transformer-based BERT model is used as the primary model to improve accuracy in detecting complex and context-dependent misinformation. The system generates classification results along with confidence scores to indicate prediction reliability. To enhance detection performance, additional modules are integrated, including source credibility analysis, writing style and behavioral pattern analysis, and external verification using trusted platforms. A decision fusion mechanism combines outputs from all modules to produce a final, more reliable prediction. The system is designed for real-time operation, allowing users to input text or URLs through a user-friendly interface and receive instant analysis results. By integrating multiple detection techniques into a unified framework, the proposed system provides a scalable, accurate, and practical solution for combating fake news and malicious content in modern digital

environments.

# *Acknowledgements*

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# Chapter 1

## Introduction

### 1.1 Evolution of Information Systems

Over the last twenty years, the internet has changed from a simple information system into a powerful and interactive communication platform. In the past, most information was created and published by trusted organizations such as newspapers, television channels, and official institutions. These organizations followed a proper process that included editing, fact-checking, and verification before publishing any information. However, with the growth of social media platforms such as Facebook, Twitter (X), YouTube, and various online blogs, this system has changed completely. Now, anyone can create and share information instantly without any checking or approval. This has made communication faster, easier, and more accessible to everyone. People can now share opinions, news, and experiences with a large audience within seconds. Although this change has many benefits, it has also created serious problems. One of the biggest problems is the spread of incorrect or misleading information, which is known as information disorder. Information disorder can be divided into three types. Misinformation is false information that is shared without the intention to harm, such as incorrect health advice. Disinformation is false information that is shared intentionally to mislead people, often for political or financial reasons. Mal information is true information that is used

in a harmful or misleading way, such as sharing private data to damage someone's reputation. Because there are no strong systems to check and filter information before it is shared, false information spreads very quickly. In many cases, fake news spreads faster than real news. For example, during health emergencies, false claims about cures or treatments spread widely on social media before being corrected by trusted sources such as the World Health Organization (<https://www.who.int>) or the Centers for Disease Control and Prevention (<https://www.cdc.gov>). This creates confusion and makes it difficult for people to trust online information.

## **1.2 Nature of Fake News**

Fake news can appear in different forms and is not always completely false. In many cases, it contains partial truth, manipulated information, or misleading context. Because of this, fake news often looks realistic and becomes difficult for users to identify. The major forms of fake news are discussed below.

### **1.2.1 Fabricated Content**

Fabricated content refers to completely false information that is created intentionally to deceive readers. This type of content has no factual basis and is often used to attract attention, create panic, or manipulate public opinion. For example, a headline such as "All banks will be permanently closed tomorrow" can spread fear among people even though it is not true.

### **1.2.2 Misleading Content**

Misleading content uses real information but presents it in a distorted or confusing way. It may include incomplete data, wrong interpretation, or outdated information shared as current news. For example, an old news article or image may be reshared as a recent event, leading people to misunderstand the situation.

### **1.2.3 Imposter Content**

Imposter content occurs when fake sources pretend to be trusted organizations or official institutions. These sources copy logos, names, or website designs of credible platforms.

For example, a fake website may look similar to trusted sources like BBC or Prothom Alo, making it difficult for users to distinguish between real and fake information.

### **1.2.4 Manipulated Content**

Manipulated content includes edited images, videos, audio, or text that are altered to mislead users. With the help of modern tools and AI technologies, such as deepfake, this type of content can appear highly realistic.

For example, a video may be edited to show a person saying something they never actually said.

## **1.3 Malicious Content in Cybersecurity**

Malicious content refers to harmful digital content that is created to deceive users, steal sensitive information, or damage systems. Unlike fake news, which mainly spreads misinformation, malicious content directly targets user security and privacy. The common types of malicious content are described below.

### **1.3.1 Phishing Emails Requesting Sensitive Data**

Phishing emails are fake messages that appear to come from trusted organizations such as banks, government agencies, or online services. These emails try to trick users into sharing sensitive information such as passwords, bank details, or verification codes.

For example, a message may say “Your account will be blocked. Verify your details now.”

### **1.3.2 Fake Job Offers or Lottery Messages**

These messages attract users by offering rewards such as jobs, money, or prizes. They often request personal information or payment from the user.

For example, “Congratulations! You have won a lottery. Send your personal details to claim the prize.”

### **1.3.3 Fraudulent Financial Schemes**

Fraudulent schemes include fake investment opportunities, donation scams, or financial offers that promise high returns in a short time. These are designed to steal money from users.

For example, a message may claim “Invest today and earn double profit within one week.”

### **1.3.4 Malware Distribution Links**

Malware links are harmful links that install malicious software on a user’s device when clicked. These links may appear in emails, SMS, or social media posts.

For example, a message may say “Download this file to check your result,” but the file may contain malware.

## **1.4 Impact of Misinformation and Malicious Content**

The effects of misinformation and malicious content are far-reaching and influence multiple aspects of modern society. These impacts are not only limited to individuals but also affect organizations, governments, and overall digital environments. The spread of false or harmful content can create serious problems in social, economic, political, and security domains. The major impacts are discussed below.

### **1.4.1 Social Impact**

Misinformation can create panic, confusion, and misunderstanding among people. When false information spreads quickly through social media platforms, users often react without verifying the facts. For example, false news about natural disasters, health emergencies, or public safety can create unnecessary fear and chaos. It also leads to the rapid spread of rumors, which can damage relationships and communities. Over time, repeated exposure to fake news reduces public trust in media, news sources, and even official institutions. As a result, people may find it difficult to distinguish between real and false information, which weakens overall awareness and decision-making.

### **1.4.2 Economic Impact**

Fake and malicious content can cause significant financial damage. Many users become victims of online scams such as phishing messages, fake job offers, or lottery fraud. These scams can lead to direct financial loss when users share sensitive information or transfer money. Businesses can also suffer due to false information spread online. Negative rumors or fake news about a company can reduce customer trust and affect sales. In addition, misinformation can create instability in financial markets by influencing investor behavior and causing sudden changes in market trends.



### 1.4.3 Political Impact

Misinformation can strongly influence political systems and democratic processes. Fake news is often used to manipulate public opinion, especially during elections. False or misleading information about political leaders or policies can affect how people think and vote. It can also be used for propaganda campaigns, where information is intentionally spread to support a particular group or agenda. This may create division among people, increase conflicts, and reduce trust in political systems. As a result, misinformation becomes a serious threat to fair and transparent governance.

### 1.4.4 Security Impact

Malicious content has a direct impact on cybersecurity and digital safety. It can lead to cyberattacks, identity theft, and unauthorized access to personal or organizational data. For example, phishing attacks can steal login credentials, while malware links can infect devices and damage systems. Identity theft is another major concern, where attackers misuse personal information for illegal activities. These threats can affect individuals, businesses, and even government systems, leading to serious security risks. As technology continues to grow, the impact of such attacks is becoming more severe and difficult to control.

## 1.5 Motivation for the Project

The problem of fake news and malicious content is even more serious in developing countries such as Bangladesh. In recent years, internet access and smartphone usage have increased rapidly, which has led to a significant rise in social media usage. Platforms like Facebook and YouTube have become the primary sources of news and information for many users. However, digital literacy and awareness about fake news and cybersecurity threats are still limited among a large portion of the population.

Because of this lack of awareness, many users are unable to properly verify the authenticity of the information they receive. They often trust and share content without checking its source or accuracy. As a result, false information spreads quickly and widely. In addition, users are more likely to fall victim to malicious activities such as phishing attacks, scam messages, and fraudulent websites, which can lead to financial loss, data theft, and privacy issues.

Although there are existing solutions such as fact-checking platforms like <https://www.snopes.com> and <https://www.factcheck.org>, these systems have certain limitations. They mainly work in a reactive manner, meaning they verify information only after it has already spread across the internet. By the time a piece of information is confirmed as false, it may have already reached a large number of users and caused significant damage. Therefore, these solutions are not effective for real-time prevention.

Furthermore, many current automated detection systems focus only on analyzing the textual content of news or messages. They often ignore other important factors that can improve detection accuracy. For example, the credibility of the source, the writing style of the content, and verification from trusted platforms are not always considered. Ignoring these factors can reduce the effectiveness of the system and increase the chances of incorrect classification.

Therefore, there is a strong need for a smart, intelligent, and real-time system that can detect fake news and malicious content instantly before it spreads widely. Such a system should not only analyze the text but also evaluate the reliability of the source, detect suspicious patterns in writing style, and verify the information using trusted platforms such as Google News or APIs like <https://newsapi.org>.

This project is motivated by the goal of developing an AI-based detection system that integrates multiple techniques, including natural language processing, machine learning, and external verification. The aim is to build a system that can provide fast, accurate, and reliable results, enabling users to identify harmful or misleading content easily. Ultimately, the system will help reduce the spread of misinformation, protect users from cyber threats, and improve overall digital safety and awareness.

## 1.6 Objective

### 1.6.1 General Objective

The primary objective of this project is to design, develop, and implement a robust, scalable, and intelligent AI-driven system that can effectively detect, analyze, and classify fake news and malicious content in real time. The system aims to address the increasing challenges of misinformation and cybersecurity threats that arise in modern digital communication environments, especially on social media platforms, messaging applications, and online news portals. This objective is not limited to basic text classification. Instead, it focuses on building a system that can understand the context, meaning, and intent of content. The system will apply advanced natural language processing techniques and machine learning models to analyze textual data in a more meaningful way. By doing so, it can differentiate between real, fake, and malicious content more accurately. In addition, the system is designed to support multi-layered analysis and verification. It will not rely only on the content itself but will also consider other important factors such as the credibility of the source, writing style patterns, and verification from trusted external platforms. This multi-dimensional approach will improve the reliability and effectiveness of the detection process. Another important goal is to ensure real-time processing capability. Since misinformation spreads very quickly, especially on social media, the system must be able to analyze and provide results instantly. This will help reduce the spread of harmful content before it reaches a large audience. The system is also intended to be scalable and adaptable. It should be capable of handling large volumes of data and continuously learning from new information. This adaptive learning ability will allow the system to improve its performance over time and remain effective even as new types of fake news and cyber threats emerge. Furthermore, the project aims to integrate multiple analytical components into a single unified framework. These components include linguistic analysis (text understanding), behavioral analysis (user interaction patterns), and contextual analysis (source

and external verification). Combining these elements will create a more comprehensive and intelligent detection system. Another key objective is to improve user trust and transparency. The system will provide not only the classification result but also a confidence score and a simple explanation of the decision. This will help users understand why a piece of content is classified as real, fake, or malicious, making the system more user-friendly and trustworthy. To summarize, the general objective of this project includes the following key goals: 1. Developing a fully automated detection system that reduces the need for manual verification 2. Ensuring real-time processing to minimize the rapid spread of misinformation 3. Achieving high accuracy and reliability using advanced machine learning and deep learning models such as BERT 4. Integrating multiple analysis techniques, including linguistic, behavioral, and contextual factors 5. Providing explainable results with confidence scores to enhance user trust and understanding 6. Building a scalable and adaptive system capable of handling large and dynamic data environments Ultimately, the system aims to contribute to digital safety, improve information authenticity, and strengthen cybersecurity resilience. This is particularly important in environments where misinformation and malicious content can have serious social, economic, and political consequences.

## **1.6.2 Specific Objectives**

### **1.6.2.1 Development of an NLP-Based Text Analysis System**

A key objective of this project is to develop a Natural Language Processing (NLP)-based analytical system that can understand and interpret textual content at multiple levels. The system will not only process text at a surface level but also analyze its structure, meaning, and intent. This includes lexical analysis for breaking text into tokens such as words and phrases, syntactic parsing to understand grammatical structure, and semantic analysis to extract meaning from sentences. In addition, the system will perform contextual understanding to identify relationships between words and phrases within a given context.

Through this approach, the system will be able to detect ambiguity in language, identify misleading or manipulative phrasing, and uncover hidden intent behind messages. This ensures that the model does not rely only on simple keyword matching but instead understands the deeper meaning of content, which is essential for detecting complex fake news and phishing messages.

#### **1.6.2.2 Implementation of Machine Learning Models for Classification**

Another important objective is to design and implement multiple machine learning models to classify content into predefined categories such as real, fake, and malicious. This involves training algorithms such as Logistic Regression, Naive Bayes, and Random Forest using labeled datasets. Feature engineering techniques will be applied to extract meaningful patterns from textual data, which will improve model performance.

The models will be evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score to ensure reliable results. By comparing multiple models, the system will be able to identify the most effective approach. The use of different algorithms also improves robustness and generalization, making the system more reliable across different datasets and real-world scenarios.

#### **1.6.2.3 Integration of Deep Learning Models (BERT and Transformer-Based Models)**

To enhance performance beyond traditional machine learning approaches, this project aims to integrate advanced deep learning models, particularly transformer-based models such as BERT. These models are capable of understanding context and capturing relationships between words in a more sophisticated way.

By using contextual embeddings, the system can interpret the meaning of words based on their usage in different sentences. It can also capture long-range dependencies in text, which is important for understanding complex statements. This allows

the system to detect subtle and nuanced forms of fake news and identify advanced phishing techniques that are difficult to detect using basic models.

#### **1.6.2.4 Source Credibility Analysis Module**

An important objective is to develop a source credibility analysis module that evaluates the reliability of the content source. This module will analyze historical data related to sources and calculate credibility scores based on their past behavior.

It will consider factors such as the frequency of false content, the reputation of the publisher, and the reliability of the domain. By identifying patterns of misinformation from specific sources, the system can provide an additional layer of validation. This improves overall accuracy and ensures that the system does not rely only on content analysis but also considers the trustworthiness of the source.

#### **1.6.2.5 External Verification Using Trusted APIs**

To further improve reliability, the system will include an external verification mechanism by integrating trusted APIs such as news aggregators and verified databases. The system will fetch related news from reliable sources and compare it with the input content.

This process will help determine whether the information is consistent across multiple trusted sources. As a result, the system can validate claims in real time, reduce false positives, and increase the overall trustworthiness of predictions.

#### **1.6.2.6 Writing Style and Behavioral Pattern Analysis**

Another objective is to analyze writing style and behavioral patterns that are commonly associated with fake or malicious content. The system will detect features such as clickbait language (e.g., “shocking”, “urgent”, “breaking”), excessive use of capital letters or punctuation, emotional or sensational tone, and repetitive or unnatural phrasing.

By analyzing these patterns, the system can identify manipulative intent, social engineering techniques, and low-quality or automated content. This adds another important layer of analysis that complements text-based classification.

#### **1.6.2.7 Real-Time Processing and Response Generation**

A critical objective of this project is to ensure real-time processing capability. The system should be able to analyze input data and provide results instantly with minimal delay.

This includes processing large volumes of data efficiently and delivering classification results immediately. Real-time detection is essential because misinformation spreads very quickly, and delayed responses can reduce the effectiveness of the system.

#### **1.6.2.8 Confidence Score Generation and Explainability**

To improve transparency and user trust, the system will generate a confidence score for each prediction. This score will indicate how certain the model is about its classification result.

In addition, the system will provide explainable outputs by highlighting important words or phrases and showing the factors that contributed to the decision, such as source credibility or writing style. This improves user understanding, system accountability, and interpretability of AI decisions.

#### **1.6.2.9 Development of a User-Friendly Interface**

The project also aims to develop a simple, intuitive, and user-friendly interface that allows users to interact with the system easily. Users will be able to input text or URLs and receive instant results.

The interface will display classification results, confidence scores, and explanations in a clear and understandable way. Technologies such as Flask or Streamlit will be

used to design the interface. This ensures accessibility for non-technical users and supports practical use in real-world scenarios.

#### **1.6.2.10 System Scalability and Adaptability**

Another objective is to ensure that the system is scalable and adaptable. The system should be capable of handling large volumes of data without performance issues.

It should also be able to learn from new data and adapt to evolving patterns of fake news and malicious content. This will be achieved through modular system design, regular model updates, and the ability to retrain models using new datasets.

#### **1.6.2.11 Ethical AI and Responsible Implementation**

Finally, the system will be developed with a strong focus on ethical AI principles. This includes avoiding bias in model predictions, ensuring fairness across different types of content, and protecting user privacy.

The system will also provide transparent decision-making processes so that users can understand how predictions are made. This ensures responsible use of AI and increases trust in the system.

## **1.7 Contributions**

This project makes significant contributions to the field of fake news and malicious content detection by addressing key limitations of existing systems and proposing a comprehensive, real-time, and multi-layered solution. The most important contribution of this project is the development of a unified detection framework that can identify both fake news and malicious content within a single system. Existing research, as identified in the problem statement, mainly focuses on either fake news detection or cybersecurity threats separately. This project bridges that gap



by combining both aspects into an integrated solution, making it more practical and applicable in real-world scenarios. Another major contribution is the use of a hybrid modeling approach, where traditional machine learning algorithms such as Logistic Regression, Naive Bayes, and Random Forest are combined with advanced deep learning models like BERT. This combination allows the system to achieve a balance between computational efficiency and contextual understanding, resulting in improved classification performance. The project also introduces a source credibility analysis module, which evaluates the reliability of information sources based on their historical behavior. Unlike many existing systems that rely only on text analysis, this approach adds an additional layer of validation, making the detection process more robust and trustworthy. In addition, the system integrates external verification mechanisms using trusted APIs and news platforms. By comparing input content with verified information from reliable sources, the system is able to reduce false positives and improve the accuracy of predictions. Another important contribution is the inclusion of writing style and behavioral pattern analysis. The system is capable of detecting clickbait language, emotional tone, and manipulation techniques commonly used in fake news and phishing attacks. This helps identify hidden intent and social engineering strategies that are often missed by traditional models. The project further contributes by implementing a real-time detection system that can analyze content instantly and provide immediate feedback. This is particularly important in modern digital environments where misinformation spreads rapidly, and delayed detection reduces effectiveness. Moreover, the system enhances transparency by providing explainable outputs. It generates confidence scores and highlights key factors that influence each prediction, allowing users to understand the reasoning behind the results. This improves user trust and system reliability. Finally, the project focuses on practical usability by developing a user-friendly interface that enables users to easily input text or URLs and receive instant analysis. This ensures that the system is not only technically effective but also accessible and useful for real-world applications. Overall, this project contributes a scalable, intelligent, and real-time solution that integrates multiple detection techniques into a single framework, addressing both misinformation and

cybersecurity challenges in a more comprehensive manner. This project makes significant contributions to the field of fake news and malicious content detection by addressing key limitations of existing systems and proposing a comprehensive, real-time, and multi-layered solution. The most important contribution of this project is the development of a unified detection framework that can identify both fake news and malicious content within a single system. Existing research, as identified in the problem statement, mainly focuses on either fake news detection or cybersecurity threats separately. This project bridges that gap by combining both aspects into an integrated solution, making it more practical and applicable in real-world scenarios. Another major contribution is the use of a hybrid modeling approach, where traditional machine learning algorithms such as Logistic Regression, Naive Bayes, and Random Forest are combined with advanced deep learning models like BERT. This combination allows the system to achieve a balance between computational efficiency and contextual understanding, resulting in improved classification performance. The project also introduces a source credibility analysis module, which evaluates the reliability of information sources based on their historical behavior. Unlike many existing systems that rely only on text analysis, this approach adds an additional layer of validation, making the detection process more robust and trustworthy. In addition, the system integrates external verification mechanisms using trusted APIs and news platforms. By comparing input content with verified information from reliable sources, the system is able to reduce false positives and improve the accuracy of predictions. Another important contribution is the inclusion of writing style and behavioral pattern analysis. The system is capable of detecting clickbait language, emotional tone, and manipulation techniques commonly used in fake news and phishing attacks. This helps identify hidden intent and social engineering strategies that are often missed by traditional models. The project further contributes by implementing a real-time detection system that can analyze content instantly and provide immediate feedback. This is particularly important in modern digital environments where misinformation spreads rapidly, and delayed detection reduces effectiveness. Moreover, the system enhances transparency by providing explainable outputs. It generates confidence scores and highlights key factors that

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## 1.8 Challenges

Detecting fake news and malicious content is a complex and challenging task due to the dynamic nature of digital communication and the evolving strategies used by attackers. One of the major challenges is understanding context and intent. Fake news often uses subtle language, sarcasm, or partial truth, which makes it difficult for systems to correctly interpret meaning using simple models. Another significant challenge is the rapidly evolving nature of misinformation. New patterns, writing styles, and manipulation techniques are constantly emerging, which require the system to be adaptable and continuously updated. Static models may fail to detect newly generated fake content or advanced phishing messages. Handling multilingual and mixed-language data is also a major issue, especially in regions like Bangladesh where content may include both Bangla and English. This increases the complexity of text processing and requires models that can understand multiple languages effectively. In addition, detecting malicious content is particularly difficult because attackers use psychological manipulation techniques such as urgency, fear, and rewards. These messages often appear normal but contain hidden intent, making them harder to identify. Another challenge is data quality and availability. Reliable labeled datasets for fake news and malicious content are limited, and existing datasets may contain noise, imbalance, or inconsistent labeling. This can negatively affect model performance. Finally, integrating multiple components such as NLP

models, source credibility analysis, writing style analysis, and external verification into a single system is technically complex. Ensuring real-time performance while maintaining accuracy and scalability adds another layer of difficulty.

## 1.9 Organization of the Report

This report is organized into several chapters to provide a clear and structured understanding of the research work. Each chapter focuses on a specific aspect of the project, ensuring logical flow and easy readability.

### **Chapter 1: Introduction**

This chapter provides an overview of the research topic. It discusses the motivation behind the project, defines the objectives, highlights the key contributions, and outlines the major challenges associated with fake news and malicious content detection. It also presents the overall structure of the report.

### **Chapter 2: Related Works**

This chapter reviews existing research and previously developed systems related to fake news detection and malicious content analysis. It also discusses the limitations of current approaches and identifies gaps in existing solutions, which justify the need for the proposed system.

### **Chapter 3: Proposed Methodologies**

This chapter describes the proposed system in detail. It explains the overall methodology, including data collection, preprocessing, feature extraction, model development using machine learning and deep learning techniques, and additional modules such as source credibility analysis, writing style analysis, and external verification.

### **Chapter 4: Implementation**

This chapter focuses on the practical implementation of the system. It describes the tools, technologies, and frameworks used to develop the system, along with details about system integration and deployment.

**Chapter 5: Experimental Results**

This chapter presents the results obtained from the system. It includes performance evaluation using different metrics such as accuracy, precision, recall, and F1-score. It also provides analysis of the results and a summary of the experimental findings.

**Chapter 6: Conclusions**

This chapter summarizes the overall research work and highlights the key outcomes of the project. It also discusses the limitations of the current system and suggests possible future improvements and extensions.

Finally, the report concludes with the Bibliography, which includes all the references and sources used throughout the research.

# Chapter 2

## Related Works

[1] *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding* — In this paper, J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova introduced BERT, a transformer-based deep learning model that significantly improved natural language understanding through bidirectional context analysis. This work contributed to the field by providing a powerful pre-trained language model that can capture deep contextual relationships between words. In their approach, the model was first pretrained on a large text corpus using tasks such as Masked Language Modeling (MLM) and Next Sentence Prediction (NSP), and then fine-tuned for specific downstream tasks like text classification. The final result showed that BERT achieved state-of-the-art performance across multiple NLP benchmarks and outperformed both traditional machine learning and earlier deep learning models.

[2] *FakeNewsNet: A Data Repository with News Content, Social Context and Spatiotemporal Information* — In this paper, K. Shu, D. Mahudeswaran, S. Wang, D. Lee, and H. Liu developed the FakeNewsNet dataset, which combines news content with social context and user interaction data. This contribution is important because it provides a more comprehensive dataset for fake news detection by including multiple dimensions of information. The workflow involved collecting data from news websites and social media platforms, integrating textual content with user behavior and metadata, and structuring it for analysis. The final result was a rich

dataset that significantly improved model performance by enabling multi-feature analysis beyond text-only approaches.

[3] *BanFakeNews: A Dataset for Detecting Fake News in Bangla* — In this paper, M. Z. Hossain, M. A. Rahman, M. S. Islam, and S. Kar focused on addressing the lack of resources for low-resource languages by developing a Bangla fake news dataset. Their contribution lies in enabling research in Bangla language processing for fake news detection. The workflow included collecting data from Bangla news sources, manually labeling the data as real or fake, and applying preprocessing techniques to structure the dataset. The final result was a reliable dataset that supported the development of machine learning and deep learning models for Bangla fake news detection.

[4] *BanglaBERT: Language Model Pretraining for Bangla* — In this paper, A. Bhattacharjee, T. Hasan, W. U. Ahmad, et al. proposed BanglaBERT, a transformer-based language model specifically designed for Bangla text. This work contributed by improving contextual understanding in low-resource languages. The authors followed a workflow where the model was pretrained on a large Bangla corpus and then fine-tuned for various NLP tasks. The final result demonstrated that BanglaBERT achieved better performance compared to multilingual models, especially in Bangla text classification tasks.

[5] *Evaluating BERT-based Language Models for Detecting Misinformation* — In this paper, R. Anggrainingsih, G. M. Hassan, and A. Datta evaluated the effectiveness of BERT-based models in detecting misinformation. Their contribution was to analyze and compare the performance of transformer-based models with traditional approaches. The workflow involved preprocessing the dataset, generating embeddings using BERT, training classification models, and evaluating their performance. The final result showed that BERT-based models significantly outperformed traditional machine learning techniques in terms of accuracy and contextual understanding.

[6] *An AI-Driven Framework for Real-Time Fake News Detection* — In this paper, U. Twaha and Y. Arfin proposed a real-time AI-based framework for detecting fake

news. Their contribution lies in designing a system capable of processing streaming data. The workflow includes continuous data collection, preprocessing, classification using machine learning models, and a feedback loop for system improvement. The final result demonstrated effective real-time detection, although challenges related to scalability and computational cost were observed.

[7] *Enhancing Fake News Detection with Transformer-Based Deep Learning* — In this paper, N. Raza et al. focused on improving fake news detection using transformer-based deep learning models. Their contribution was to show the effectiveness of transformer architectures in capturing contextual information. The workflow involved preprocessing the dataset, generating contextual embeddings using transformer models, and applying classification techniques. The final result showed higher accuracy compared to traditional CNN and RNN-based approaches.

[8] *Fake News Detection Using Machine Learning and BERT: A Comparative Study* — In this paper, A. Nayak and M. Navalan compared traditional machine learning models with BERT-based models. Their contribution was to highlight the strengths and weaknesses of both approaches. The workflow included applying TF-IDF feature extraction for machine learning models and using BERT for contextual embedding, followed by performance comparison. The final result indicated that BERT provides higher accuracy, while traditional models are faster but less effective in handling complex data.

[9] *Multimodal Fake News Detection Using Self-Learning Framework* — In this paper, H. Chen et al. introduced a multimodal fake news detection system that analyzes text, images, and videos. Their contribution was to improve detection accuracy by combining multiple data sources. The workflow included extracting features from different modalities, integrating them using a fusion mechanism, and performing classification. The final result showed improved accuracy, especially in detecting fake news involving manipulated visual content.

[10] *Hybrid RoBERTa and XGBoost Model for Fake News Detection* — In this paper, A. Ali et al. proposed a hybrid model that combines deep learning with traditional machine learning techniques. Their contribution was to improve robustness and



performance by integrating multiple approaches. The workflow involved generating contextual embeddings using RoBERTa, followed by classification using XGBoost along with metadata features. The final result achieved state-of-the-art performance in fake news detection tasks.

[11] *BERT-LSTM Approach for Misinformation Detection* — In this paper, J. Wang, X. Wang, and A. Yu proposed a hybrid model combining BERT and LSTM. Their contribution was to enhance both contextual and sequential understanding of text. The workflow included extracting embeddings using BERT, feeding them into an LSTM network for sequence modeling, and performing classification. The final result showed improved early-stage detection and higher accuracy compared to standalone models.

# Chapter 3

## Proposed Methodologies

### 3.1 Introduction

### 3.2 System Model and Assumptions

### 3.3 Proposed Method

#### 3.3.1 Any Secondary Contribution

#### 3.3.2 Discussion

### 3.4 Conclusion

# Chapter 4

## Performance Evaluation

### 4.1 Simulation or Testbed Environment

### 4.2 Performance Metrics

### 4.3 Results Analysis

### 4.4 Conclusion

# Chapter 5

## Conclusions

### 5.1 Summary of Work

### 5.2 Future Direction

# Bibliography

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